

GRACE ONESIME ZONGO

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Electrical Engineering student with CS minor (expected May 2027) with three internships across embedded systems, HVAC controls, and power electronics. Experienced with bare-metal C/C++, PCB layout in Altium/KiCad, and FPGA design in Verilog.

EDUCATION

University of Wisconsin, Madison

September 2024 - May 2027 (expected)

- **Major:** B.S. Electrical Engineering & **Minor:** Certificate in Computer Science
- UW Credit Union Scholar and JCKF scholar and Wisconsin Idea Fellow

Madison College, Madison, WI

January 2022 – May 2024

- Associate degree: Liberal Arts Transfer: Science / Math / Technology

TECHNICAL SKILLS

Programming: C/C++, Python, Java, Verilog, MATLAB, Assembly

Tools/Hardware: Altium, KiCad, LTspice, Intel Quartus (FPGA), Oscilloscope, DMM, signal Generator, Git/GitHub.

Concepts and protocols: Embedded Systems, Power Electronics, Low-Power Design, UART/USART, I²C, SPI, SDI-12, CAN

INTERNSHIP EXPERIENCES

Firmware / AWS Student Intern

Space Science and Engineering Center – UW Madison | Madison, WI

April 2024 – Present

- Reduced system power consumption by 78% and improved runtime efficiency by 85% by implementing a custom low-power sleep mode and C++ Finite State Machine, and Integrated multiple sensors over SPI/I2C/SDI-12/Frequency.
- Co-authored a WAMC 2026 presentation (NSF-supported). The station ran unattended in Antarctica for a full year.
- Implemented on-board data logging for on-site debugging, and working on implementing a self-diagnostic system

Design Engineer Intern

Alpha Controls and Services | Middleton, WI

May 2025 – December 2025

- Analyzed HVAC mechanical blueprints to perform material takeoffs, select sensors, actuators, and Schneider controllers.
- Created electrical schematics and wiring diagrams for HVAC systems using equipment submittals and sequence of operation.
- Prepared cost estimates for projects by quantifying control system components and materials.

Electrical Engineering Intern

Bemis Manufacturing Company | Madison, WI

June 2022 – August 2022

- Designed a prototype of Bemis's first smart, health monitoring bidet seat for household use.
- Integrated an optical heart-rate sensor into the seat, with a mechanism to record its health data.

PROJECT EXPERIENCE

Digitally Controlled Buck Converter (personal project)

2026

- Built a synchronous DC-DC buck converter with a UART-programmable 1.8–10 V output. Bare-metal C on STM32G431KB with 200 kHz PWM, ADC sensing, and a 10 kHz PID loop with anti-windup and overcurrent protection.
- Laid out the PCB in Altium, verified the power stage in LTspice, and debugged bring-up faults on the bench.

Promoting Electric Propulsion (PEP) competition – Electric Catamaran

September 2023 – April 2024

- Designed a dual 18650 Li-ion battery system and differential thrust control for an electric catamaran, contributing to a 2nd place finish at the PEP 2024 Competition.

AM Radio Receiver

September 2023 – April 2024

- Designed and assembled a 4-stage BJT AM receiver with RF amplification, envelop detector, Audio amplification.
- Verified each stage on a bench using a signal generator and oscilloscope and achieved audio reproduction.

Honors Project: 8-bit CPU on FPGA

June 2023 – December 2023

- Designed and simulated a Von Neumann 8-bit CPU: 8-bit register, RAM, ALU, and instructions from scratch using Quartus.
- Implemented FETCH, LOAD, and ADD instructions, and designed binary to BCD converter for segment displays.

LEADERSHIP EXPERIENCE

Wisconsin Idea Fellow | Tech'Up Jeunes Initiative | UW–Madison

May 2026 – May 2027

- Design and lead a hands-on electronics program for high-school students in Burkina Faso with Leading Change Africa.

Clubs Development Coordinator | Executive Leadership Team (ELT) | Madison College

March 2022 – May 2024

- Supported 50+ student clubs, administered the campus-wide club platform, and organized officer training.